EART26201 | Principles of Geochemistry | Practical 2

Isochron dating and isotopic tracers of magma sources

FEEDBACK

Task

This practical contains two questions about using radiogenic isotopes to date meteorite formation and track crustal contamination recorded by lavas erupted from an Andean volcano. Each question should take about 60 minutes.

I strongly encourage you to discuss your work with each other as you go.

Demonstrators will be available throughout the practical class, and I will provide a short round-up at the end of the session. Feedback will be posted on Blackboard in due course.

Please share any comments you have as we go along, and I will do my best to adjust.

I suggest typing your answers into this document and/or its associated Excel file.

Goals

At the end of this practical you should be able to:

- Plot isochron diagrams in Excel
- Interpret isochron diagrams to determine ages of geological samples
- Interpret magmatic radiogenic isotope data in terms of different magma sources
- Calculate mixing trends between different isotopic reservoirs

Other themes relevant to this week's content are reviewed in a Blackboard quiz.

You will revisit many of these themes in EART22200 (Igneous Minerals and Processes) and EART24200 (Metamorphic Minerals and Processes) later in the year.

Before you start

Open the file *EART26201_Practical2.xlsx* in Excel (or Libre Office Calc or similar).

Question 1: Meteorite crystallisation ages (60 minutes)

a) Juvinas and Nakhla are both achondrite meteorites. Neither have chondrules and both have broadly basaltic textures and compositions: Juvinas contains pyroxene and plagioclase while Nakhla contains olivine, pyroxene and plagioclase. Briefly describe how achondrite meteorites form and how they are related to chondritic meteorites.

Achondrites are silicate meteorites from the crusts and mantles of differentiated parent bodies. That is, they are sourced from asteroids that have differentiated into metal-rich cores and silicate-rich crusts and mantles. In contrast, chondritic meteorites are from undifferentiated parent bodies.

- Remember that the radiogenic isotope evolution of a sample can be described using the following fundamental equation of geochronology:

$$\frac{D}{S} = \left(\frac{D}{S}\right)_0 + \frac{N}{S}\left(e^{\lambda t} - 1\right)$$

This equation is expressed as follows when applied to the Rb-Sr system:

$$\frac{^{87}\text{Sr}}{^{86}\text{Sr}} = \left(\frac{^{87}\text{Sr}}{^{86}\text{Sr}}\right)_0 + \frac{^{87}\text{Rb}}{^{86}\text{Sr}}\left(e^{\lambda t} - 1\right)$$

- The Rb-Sr systematics of different phases in Juvinas are given in Table 1 below:

Table 1: Rb-Sr systematics of the Juvinas meteorite.

Sample	**^{87}Sr/^{86}Sr**	**^{87}Rb/^{86}Sr**
Glass	0.7049	0.090
Tridymite/quartz	0.7006	0.024
Pyroxene	0.6996	0.009
Whole-rock	0.6994	0.006
Feldspar	0.6993	0.005

b) Plot an isochron diagram (^{87}Rb/^{86}Sr on x-axis and versus ^{87}Sr/^{86}Sr on y-axis) for Juvinas using the data in Table 1 (*Insert* → *Charts* → *Scatter*). You can do this in the sheet titled *Question 1* in the Excel file you have downloaded. (Check you can copy from Word to Excel – a major source or errors by student and academic alike!).

See Excel spreadsheet

c) Perform a simple linear regression by clicking on *Add Chart Element* → *Trendline* → *Linear.* Display the equation on the on the isochron diagram by right clicking on the trendline and then selecting Format *Trendline* → *Display Equation on Chart*. Can you

obtain a good fit to the data, implying that the system has remained closed and was in isotopic equilibrium at $t = 0$?

d) Given that $\lambda_{(^{87}\mathrm{Rb})} = 1.42 \times 10^{-11}$ years, how old is Juvinas?

See Excel spreadsheet for working. According to Sr isotopes, Juvinas formed at 4.47 Ga

e) What $(^{87}Sr/^{86}Sr)_0$ content is recorded by Juvinas?

0.699

- The Rb-Sr systematics of different phases in Nakhla are given in Table 2 below:

Table 2: Rb-Sr systematics of the Nakhla meteorite.

Sample	$^{87}Sr/^{86}Sr$	$^{87}Rb/^{86}Sr$
Plagioclase A	0.70492	0.1383
Diopside D	0.70421	0.0937
Diopside G	0.70344	0.0522
Diopside H	0.70469	0.1232

f) How old is Nakhla?

See Excel spreadsheet for working. According to Sr isotopes, Nakhla formed at 1.20 Ga

g) What $(^{87}Sr/^{86}Sr)_0$ content is recorded by Nakhla?

0.7026

h) Rb-Sr systematics are sometimes affected by alteration processes that occur on other planetary bodies as well as on Earth. To make sure that the age you calculated for Nakhla using the Rb-Sr system is correct, plot an isochron diagram using Sm-Nd systematics provided in Table 3 below knowing that $\lambda_{(^{147}\mathrm{Sm})} = 6.54 \times 10^{-12}$ years:

Table 3: Sm-Nd systematics of the Nakhla meteorite.

Sample	$^{143}Nd/^{144}Nd$	$^{147}Sm/^{144}Nd$
Whole-rock 1	0.512628	0.09887
Whole-rock 2	0.512660	0.10284
Pyroxene 1	0.513093	0.15054
Pyroxene 2	0.513058	0.15209

See Excel spreadsheet. According to Nd isotopes, Nakhla formed at 1.29 Ga – the Sr and Nd ages are broadly coherent and reliable.

i) Did Juvinas and Nakhla form at the same time? If not, why might the meteorites have different ages? What does this tell us about where they formed? (You will need to write at least a couple of sentences to fully answer this question)

(Clue, exposure dating of the surface of the Nakhla meteorite suggest that it spent only 11 Ma in space before falling to Earth)

Juvinas formed very early in the history of the solar system, not long after the condensation of the solar nebula. It was likely ejected from its parent body shortly after differentiation had occurred. In contrast, Nakhla formed relatively recently and must have formed though magmatic activity on a recently active planetary body (i.e. a terrestrial planet). Indeed, its short exposure age suggests that it was ejected into space by a very recent impact. In fact, Nakhla actually forms part of the SNC group of meteorites that are thought to have been sourced from Mars.

Question 2: Tracking crustal contamination at Cerro Galán, Argentina (60 minutes)

- Cerro Galán is a large calc-alkaline volcano in the high Altiplano of the Argentinian Andes. Magmas erupted from arc volcanoes like Cerro Galán are often complex mixtures of basaltic melts from the mantle and dacitic/rhyolitic melts from the continental crust. Isotopic tracers can be used to determine the relative contributions of mantle-derived basalts and crust-derived dacites/rhyolites to erupted magmas.



Figure 1: Inside the Cerro Galán caldera. credit: sebastian del val.

- The sheet titled *Question 2* contains a data table with Sr concentrations and Rb-Sr systematics of samples from Cerro Galán published by Francis *et al.* (1980).

a) Plot an isochron diagram for these samples. Is it possible to calculate a unique age for these samples? If not, why not?

See Excel spreadsheet. It is not possible to plot an isochron diagram for these samples. They do not form on a single line, implying that they have different $(^{87}Sr/^{86}Sr)_0$ and therefore different sources. Thus no isochron can be fitted to determine a unique age.

b) Combine trace element and isotope data on a new plot with Sr concentrations on the x-axis and $^{87}Sr/^{86}Sr$ contents on the y-axis (you will need to be careful about how you select the data columns for this). Knowing that mantle-derived melts almost always have lower $^{87}Sr/^{86}Sr$ contents than the continental crust, which sample is most likely to represent mantle melts supplied to base of the Andean crust?

G49 has the lowest $^{87}Sr/^{86}Sr$ and is therefore the best candidate for a mantle melt

c) Calculate the composition of melts produced by mixing the low-$^{87}Sr/^{86}Sr$ mantle-derived reservoir identified in b) with a high-$^{87}Sr/^{86}Sr$ crustal reservoir containing 150 µg/g Sr with an $^{87}Sr/^{86}Sr$ of 0.718. Do this by adding crustal melts to the mantle-derived melt in increments of 10 wt.% (i.e., mix crustal and mantle-derived melts in proportions of 0 and

1, then 0.1 and 0.9 and so on). Note that mixing trends on isotope plots are curved because the mixing end-members contain different amounts of Sr (contrast this with the linear mixing trends calculated for elemental data in Practical 1). Before working out the Sr isotope contents of mixed magmas, you must first calculate the amount of Sr contributed from each reservoir at each mixing increment.

There are some hints in the Excel file to guide you, but if you're still struggling after a while, ask a demonstrator for help.

See Excel spreadsheet

d) How much crustal contamination is needed to explain the compositions of G26 and G55?

G26 can be explained by the addition of 80 wt.% crust to the mantle-like composition of G49; G55 can be explained by the addition of 40 wt.% crust.

e) How well does the mixing trend you've calculated describe the compositions of samples from Cerro Galán overall? How important is crustal contamination at this volcano?

The simple mixing trend calculated here explains the compositions extremely well; most data points lie on or close to the predicted trend. Crustal contamination is very important at Cerro Galán with most samples recording >50 wt.% contamination.

f) What are the implications of your mixing model for the Sr isotope content of the mantle beneath Cerro Galán? Is there any evidence for Sr isotope variability?

Samples lie very close to a single mixing line, implying that there is very little variability in the Sr isotope content of melts supplied from the mantle. This means that the mantle must either be very homogenous, or that any mantle-derived heterogeneity is erased by mixing before even reaching the continental crust.

g) Crystals erupted from volcanoes like Cerro Galán are often compositionally zoned. That is, they often have concentric bands with different compositions, somewhat like tree rings. Figure 2 shows Sr concentrations (black squares) and $^{87}Sr/^{86}Sr$ contents (grey boxes) measured in a profile across a single crystal by Ginibre and Davidson (2014). The x-axis shows distance through the crystal. Based on what you've learnt in this practical, explain whether the outside of the crystal is at location A or location B.

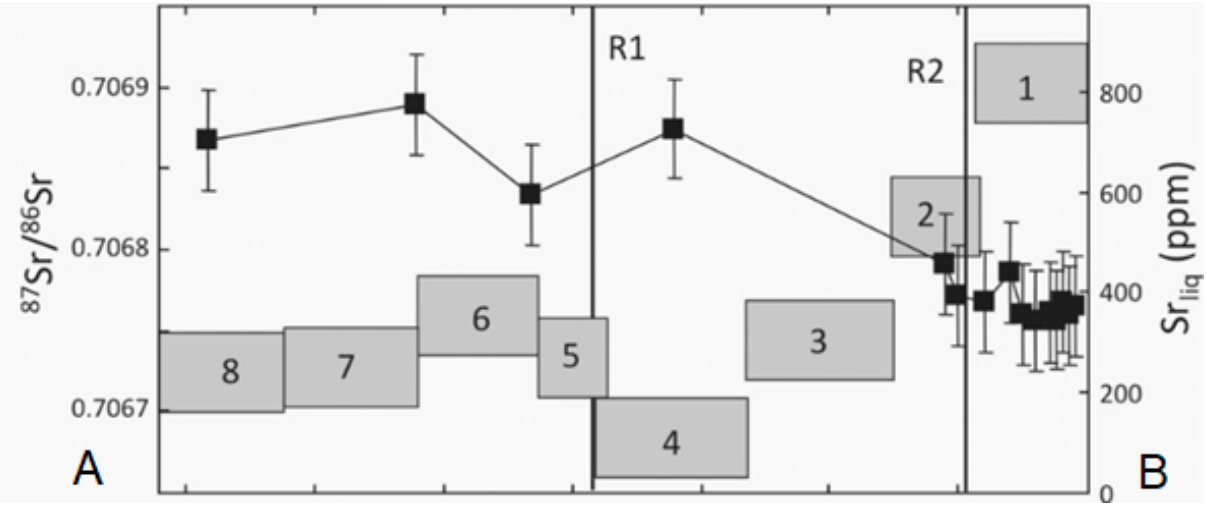


Figure 2: Sr concentrations (black squares) and $^{87}Sr/^{86}Sr$ contents (grey boxes) across a crystal from Parinacota volcano in Chile (Ginibre and Davidson, 2014)

The outside (rim) of the crystal is at location B and the inside (core) is at location A. This is because the Sr concentration decreases and the Sr isotope content increases from A to B, which is consistent from crystallisation in a mantle-derived magma that is progressively contaminated by crust with low Sr and high $^{87}Sr/^{86}Sr$.